

Towards the construction of a formal definition-classification system for recursion

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Abstract: Recursion in language is a hot topic in many fields like theoretical linguistics, psycholinguistics and neurolinguistics. However, there is no clear or consistent definition of recursion in previous studies. The current paper means to construct a formal definition-classification system for recursion in language, consisting of different subtypes of recursive rules and various models and levels of recursive sequences, exemplified with rules and sequences in English, Mandarin Chinese, etc. By doing so, the paper tries to unify some diverging definitions in the literature into a consistent system and alleviate some confusion that exists in the field.

Keywords: definition; classification; recursion, category recursion, relative clauses

1. Introduction

That “recursion is the essence of human natural language and distinguishes natural language from other forms of animal communication” has been “a continuing theme of Chomsky’s work since his 1957 book, *Syntactic Structures*, and reiterated in Hauser et al. (2002)” (Corballis, 2014: 27). In Hauser et al.’s (2002: 1569) own words, “FLN [the faculty of language in the narrow sense] only includes recursion and is the only uniquely human component of the faculty of language”. This theme is called “recursion-only hypothesis” (ROH) in Jackendoff & Pinker (2005: 212). ROH has been approved by some researchers (Premack, 2004; Fitch & Hauser, 2004; Nevins et al., 2009a, b; Friederici, 2014; Fukui, 2017) but contested by others (Everett, 2005, 2009; Jackendoff & Pinker, 2005; Evans & Levinson, 2009; Gervain et al., 2008; Hochmann et al., 2008; Corballis, 2014; Taraban & Bandara, 2017). This disputation might result from the lack of a clear or consistent definition of recursion, which has drawn criticisms in the literature, as exemplified in the following.

Dékány (2019: 319-320) mentions “the existence of (at least) two different meanings of recursion in previous studies”. One is “grammatical recursion” fueled by the Merge operation; for example, A (e.g., *the*) and B (e.g., *cat*) are merged into C (e.g., *the cat*), and C (e.g., *the cat*) and D (e.g., *meows*) into E (e.g., *the cat meows*), and so on. The other is “category recursion”, which means the “embedding of a category inside another of the same category”; for example, in *Watson thinks that Holmes loves Irene*, one clause is embedded within another clause. The latter, i.e., “category recursion”, are discussed more frequently in the literature on recursion. For example, Martins & Fitch (2014: 17-19) claim that “recursion has been used to characterize the process of embedding a constituent [of a certain kind of category] inside another constituent of the same kind”, which “generates hierarchical structures that display similar properties across different levels of embedding”, and that “this feature, called self-similarity, is a signature of recursive structures”. To illustrate this, they hold that the compound noun *[[student] committee]*, which is in a pattern of “[[[ALPHA]ALPHA]...]”, is recursive, in which “a noun phrase is nested [i.e., “embedded”] inside another noun phrase”, while a sentence with a noun and a verb together like *[[trees] grow]*, which is in a pattern of “[[[A]B]...]”, is non-recursive, wherein “a constituent of a given type [of category] is not embedded within a constituent of that same type”. Also, when investigating child acquisition

of recursive sequences, most literature (Bryant, 2006; Hollebrandse et al., 2008; Limbach & Adone, 2010; Kinsella, 2010; Pérez-Leroux et al., 2012; Hollebrandse & Roeper, 2014; Terunuma et al., 2017; Pérez-Leroux et al., 2018; Bejar et al., 2018; Li et al., 2020) only deals with “category recursion” sequences, such as “recursive possessives” (e.g., *Elmo’s sister’s ball*), “recursive prepositional phrases” (e.g., *the baby with the woman with the flowers*), “recursive adjectives/modifiers” (e.g., *(the/a) second green ball; big red balls*), “recursive locatives” (e.g., *tsukue-no osara-no ringo* ‘Lit: table-Locative plate-Locative apple; an apple on the plate on the table’), “recursive verbal nouns” (e.g., *tea pourer maker*), “recursive relative clauses (RCs)” (e.g., *the lion that is next to the bear that is next to the zebra*), and “recursive complements” (e.g., *I think you said they gonna be warm*). Recently, Tom Roeper confirms “a distinction between ‘category recursion’ and Merge”; however, Noam Chomsky holds that “‘category recursion’ is a special case of Merge” (personal communication between Tom Roeper and Noam Chomsky in 2021). Besides, Sevcenco et al. (2015: 297) hold that “recursion is evident in the basic operation of Merge at the two word stage (Hauser et al. 2002)”, while Roeper (2011: 58) claims that “Merge is a recursive operation that is invoked as soon as more than two words are combined”. Moreover, conjunction/coordination, “a simple flat serial concatenation” (Lowenthal, 2014: 210), “not a hierarchical tree-based representation” (Rohrmeier et al., 2014: 72), exemplified with the expressions in (1–4), is thought to be a (finite-state) model of recursion (Chomsky 1957, 1959; Fitch & Hauser, 2004), called “Tail Recursion” in Rohrmeier et al. (2014) and “Direct Recursion” in Roeper (2011) (conjunction is however called “iteration”, not “recursion”, in Martins & Fitch 2014). Roeper (2011) discussed two types of recursion: “Direct Recursion”, which is also called (flat) coordination/concathenation/conjunction, and “Indirect Recursion”, which is not flat but embedded/hierarchical, also called “category recursion” as mentioned above.

- (1) (The) old old old...(man comes.) (Chomsky 1959: 21)
- (2) (The) green, red, yellow, purple ... (balls are in the box.) (Rohrmeier et al., 2014: 73)
- (3) (This sentence) continues and continues and continues and continues. (*ibid*)
- (4) (he is pulling) an Elmo, a bicycle and a doll (Pérez-Leroux et al., 2012: 306)

The above controversies show the lack of general consensus as to the definition of recursion in the literature. In fact, according to Tomalin (2007, 2011), since recursion is borrowed from mathematics during the 1940s and 1950s, at least 9 possibilities can be identified based on the originating literature and current research trends. This situation has drawn many criticisms. For example, Lowenthal (2014: V) claims that previous researchers “never specify which exact definition of ‘recursion’ they use”; and Martins & Fitch (2014: 17) hold that “the diversity of definitions [of recursion] in use has prevented the consistent interpretation of empirical results”.

The field is in dire need of a clear and consistent definition of recursion. The main purpose of this study is to construct a formal definition-classification system for recursion, under which there exist different subtypes of recursive rules, and various subtypes, models and levels of recursive sequences, hopefully to unify some previous diverging definitions into a consistent system and alleviate the confusion that exists in the field.

2. The construction of a definition-classification system for recursion

This section deals with the construction of a formal definition-classification system for recursion which mainly consists of the recursive nature of Merge, the concept of the “levels” of recursive

sequences, and the potential expanding directions of recursive rules, which can classify recursive rules into subtypes like tail-, nested- and mixed-recursion rules. Moreover, inspired by Chomsky's (1957: 22) claim that besides conjunction, the finite state model of recursive sequence, "we can find various kinds of non-finite state models [of recursive sequences] within [natural languages like] English", the current paper tries to make clearer what these various kinds of models could be and list various exemplifications in natural languages like English and Mandarin, within the framework of the Minimalist Program (MP) (Chomsky, 1995, 2001, 2005, 2015, 2019).

2.1 Recursion *per se* as Merge (the recursive nature of Merge)

Within MP, recursion is also called Merge, represented as the formal equation in (5) in Fukui (2017).

(5) $K = \{\gamma, \{\alpha, \beta\}\}$, where $\gamma \in \{\alpha, \beta\}$ (The equation in (5) states that a new object K is formed by merging two objects α and β , and specifies one of them (γ) as the “head”/“label” of K .)
(Fukui, 2017: 15)

Here in (6) is how the equation in (5) restated as the two tree diagrams in a bottom-up pattern: merge α and β , projecting one of them as the head/label of the new, larger unit.

(6)

α
 $\swarrow \quad \searrow$
 $\alpha \qquad \beta$

Or:

β
 $\swarrow \quad \searrow$
 $\alpha \qquad \beta$

In a top-down pattern, the equation in (5) is restated as the two rules in (7), in which the symbol “ $\rightarrow$ ” means “is rewritten with”, which is frequently used in the earlier literature like Chomsky (1957, 1959) and recent literature like Rohrmeier et al. (2014). The two rules in (7) mean that α is rewritten with itself and β , and β is rewritten with itself and α , respectively. A rule is recursive wherein the non-terminal symbol is rewritten with itself and the other (terminal or non-terminal) symbol (cf. Rohrmeier et al., 2014). Thus, rules as in (7) are recursive.

(7) $\alpha \rightarrow \{\alpha \beta\}$; or: $\beta \rightarrow \{\alpha \beta\}$

The unorderedness of the two elements in the curly brackets “{ }” in (5–7) means that the syntactic operation Merge does not specify the linear order of the two objects in the curly brackets (Chomsky 2005 holds that the linear order differences between languages are in the phonological component instead of the syntactic component). In order to show the fixed linear orders of actual expressions/sequences in a certain language, say, English or Mandarin, the rules in (7) are further restated as the ones in (8) (angle brackets contain ordered elements).

(8) $\alpha \rightarrow \langle \alpha \beta \rangle$; or: $\alpha \rightarrow \langle \beta \alpha \rangle$; or: $\beta \rightarrow \langle \alpha \beta \rangle$; or: $\beta \rightarrow \langle \beta \alpha \rangle$

That the rules in (5–8) are recursive means that these rules can be used for potentially infinite times if necessary during the generative processes of recursive sequences. For example, the α in the brackets of “ $\alpha \rightarrow \langle \alpha \beta \rangle$ ” in (8) can be further rewritten with $\langle \alpha \beta \rangle$, resulting in the rule in (9), wherein the α in the brackets can be further rewritten with $\langle \alpha \beta \rangle$, and so on, unless at some stage it is stopped by the combination with recursion-terminating rules like the ones in (10a–b) in order to generate recursive sequences (there are only terminals in a sequence, which cannot be further rewritten).

(9) $\alpha \rightarrow \langle \langle \alpha \ \beta \rangle \ \beta \rangle$

(10) a. $\alpha \rightarrow T$ b. $\beta \rightarrow K$

(11) $\beta \rightarrow \langle \alpha \langle \alpha \beta \rangle \rangle$

Similarly, the β in the brackets of “ $\beta \rightarrow \langle \alpha \beta \rangle$ ” in (8) can be rewritten with $\langle \alpha \beta \rangle$, resulting in the rule in (11), wherein the β in the brackets can be further rewritten with $\langle \alpha \beta \rangle$, and so on, unless at some stage it is stopped by the involvement of the rules as in (10a–b) in order to generate recursive sequences. The same is true with “ $\alpha \rightarrow \langle \beta \alpha \rangle$ ” and “ $\beta \rightarrow \langle \beta \alpha \rangle$ ” in (8). In other words, each of the rules in (8) can be used once, twice..., and so on, unless at some stage the new rule is combined with recursion-terminating rules as in (10) to get sequences. If a rule in (8), say “ $\alpha \rightarrow \langle \alpha \beta \rangle$ ”, is used once and then combines with the rules in (10), the 1-level recursive sequence TK is generated; if it is used twice and then combines with the rules in (10), the 2-level recursive sequence TKK is generated, To note, the recursive sequences like TKK, derived from the combination of the rule in (9) and those in (10), do not involve the coordination/conjunction of the Ks. Instead, these Ks are in a hierarchical relationship (cf. Rohrmeier et al. 2014: 73.)

Also to note, although “the ‘head’/‘label’ of the K” is mentioned in (5) above, “Chomsky (2001, 2005, etc.) concludes that [as for natural languages,] the notion ‘labels’ may be illusory with no theoretical status and can be eliminated by virtue of the third-factors such as minimal search conditions” [the third factors are “principles not specific to the faculty of language” (Chomsky 2005: 6)]. This is because in natural language structures of the form {H XP}¹, “representative cases of Merge”, “minimal search conditions trivially determine that Head (H) is the element that enters into further computations” and XP “has no information available with its head too deep down within it” (Fukui, 2017: 16). In a word, in natural language, H and XP are merged into a larger unit which is naturally and automatically “headed” by H (there is no need to make a special effort to “label”). In other words, H is naturally and automatically rewritten with H and XP, clearly showing the recursive nature of Merge in natural language. Moreover, under the full “Bare Phrase Structure” theory (an important part of the MP; Chomsky 2001, 2005, 2019), H and X (not XP) are merged into a unit headed by H. That is, H is rewritten with H and X (not XP). The current paper adopts this full “Bare Phrase Structure” theory in the following analyses, for it states the recursive nature of Merge in natural language in a consistent pattern. For example, Verb (V) and Noun (N) (not NP) are merged into a new unit headed (not labeled) by V, forming the recursive rule in (12a). The rule in (12a) is recursive in that the symbol V is rewritten with itself and the other symbol N, so that (potentially) (12a) can lead to (12b) and so on if interpretable at the “syntax-semantics interface” (Chomsky 2005). Similarly, Adjective (A) and N are merged into a new unit headed by N, forming the recursive rule in (13a), which can lead to (13b) and so on if interpretable at the syntax-semantics interface. Moreover, the rules in (12a) and (13a) can be combined together to get a new recursive rule “V→<V <A N>>”. Dékány’s (2019) “grammatical recursion” outlined in Section 1 above, which is also called “Merge”, is in fact distinct from “Merge” defined here, which takes “headedness” (the recursive nature of Merge) into consideration, in line with Chomsky (2001, 2005, etc.), Fukui (2017), and Freidin (2014: 144-145), who summarizes that “the formulation of Merge is arguably the minimal specification of a recursive operation” and “Merge provides the mechanism for generating recursive structures”.

(12) a. $V \rightarrow \langle V N \rangle$ b. $V \rightarrow \langle \langle V N \rangle N \rangle$ (e.g., $\text{give} \rightarrow \langle \langle \text{give me} \rangle \text{books} \rangle$)

¹ The English letters or abbreviations in this paper mainly represent lexical categories. For instance, V stands for verb, A for adjective, N for noun, I for Inflection, D for determiner, Poss for possessive, Loc for locative, and P for prepositions. But H stands for Head and X represents a certain lexical category and the other letters like M, T, W and K represent non-terminal symbols in artificial grammar.

- (13) a. $N \rightarrow \langle A N \rangle$ b. $N \rightarrow \langle A \langle A N \rangle \rangle$ (e.g., ball $\rightarrow$ $\langle$ big $\langle$ wooden ball $\rangle \rangle$)

It is also noteworthy that, although conjunction with the formula in (14) [see (1–4) in Section 1 for examples in natural language] is called a finite-state model of “recursion” in some literature like Chomsky (1957, 1959) and Fitch & Hauser (2004), and “Direct Recursion” in Roeper (2011), it is distinct from recursion/Merge defined here, since Merge here is binary-branching and must be “headed” with one of the two constituents put together.

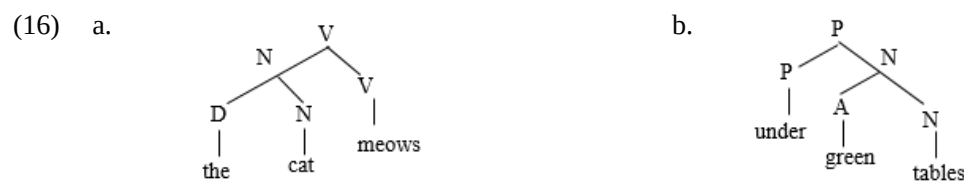
- (14) $k = x + (\text{and}) + y + (\text{and}) z \dots$ (k is a new unit and x, y, z , etc. are of the same nature).

2.2 To reconsider some (short or long) expressions as one-level recursive sequences

Under the formal definition for recursion *per se* and also for the “levels” of recursive sequences, presented in Section 2.1, the English sequence *the cat meows* mentioned in Dékány (2019) (see Section 1) is reconsidered here as a 1-level recursive sequence, generated by combining the recursive rules in (15a) and (15b), into the rule in (15c)², which further combines with the recursion-terminating rules in (15d–f).³

- (15) a. $N \rightarrow D N$ b. $V \rightarrow N V$ c. $V \rightarrow D N V$
 d. $D \rightarrow \text{the}$ e. $N \rightarrow \text{cat}$ f. $V \rightarrow \text{meows}$

In a bottom-up pattern, here in (16a) is how *the cat meows* is constructed in the tree diagram: first, merge the determiner/D *the* and the noun/N *cat*, projecting N as the head of the new unit *the cat*; second, merge the verb/V *meows* and the unit *the cat*, projecting V as the head of the new, larger unit *the cat meows*. Similarly, (16b) shows how the 1-level recursive sequence *under green tables* is constructed. There is no rule used twice in either of them.



For another example, the recursive rule in (17), derived from the combination of the recursive rules in (18a–e), is long. With the help of terminal symbols (i.e., words), it can generate a long 1-level recursive sequence as exemplified in (19). In its derivation, no recursive rule is repeatedly used, and thus it is still 1-level, in spite of its length.

- (17) $I \rightarrow N [I [V [D [A N]]]]$
 (18) a. $N \rightarrow A N$ b. $N \rightarrow D N$ c. $V \rightarrow V N$ d. $I \rightarrow I V$ e. $I \rightarrow N I$
 (19) Rose can eat a big apple

To note, the controversies mentioned in Section 1 between Sevcenco et al.’s (2015) claim that recursion is evident in Merge at the two-word stage, and Roeper’s (2011) claim that recursion is evident in Merge at the three-word stage, can be solved under the definition system in the current paper. To be specific, a two-word sequence like *trees grow* is a 1-level recursive sequence (cf.

² For natural language, “{ }” or “< >” is omitted in that the linear order of sequences is usually fixed.

³ In English, the rule “ $V \rightarrow N V$ ” cannot be used twice; that is, there is no 2-level recursive sequence corresponding to “ $N [N V]$ ”. However, in Chinese, it can; for example, the expression in (1) is a typical Chinese topic structure (Xu, 2006).

(1) [zhe-wei laofuren, [ta-erzi qu-le yi-wei meiren]]
 this-CL old-lady her-son marry-ASP one-CL beauty
 ‘Lit.: [this old lady [her son marry a beauty]]; as for this old lady, her son married a beauty.’

Martins & Fitch 2014), and a sequence including three or more than three words may still be a 1-level recursive sequence like the ones in (16) and (19); however, three-word sequences like the so-called “recursive adjectives/modifiers” (*the/a second green ball* and *big red apples* are 2-level recursive sequences (which will be explained in Section 2.3); besides, Martins & Fitch’s (2014) “recursive” sequences exemplified with two-word compound nouns like *student committee* as outlined in Section 1 are reconsidered here as 1-level recursive sequences, just like two-word expressions like *trees grow*; the only difference is that as for *student committee*, the X in the representative form {H X} of Merge happens to be of the same category as the H.

2.3 To reanalyze some “category-recursion” sequences as 2-level tail-recursion sequences

The instances of “category recursion”/“Indirect Recursion” are “a special case of Merge” in Chomsky’s opinion (see Section 1). We agree with Chomsky. However, to be more precise, “recursive adjectives/modifiers” as in (20), “recursive nouns” as in (21a–b), “recursive possessives” as in (22a–b), “recursive locatives” as in (23a–b), and “recursive prepositional phrases” as in (24) will be reanalyzed as various models of 2-level tail-recursion sequences. “Recursive RCs” as in (25) will also be reanalyzed as one model of 2-level tail-recursion sequences in Section 2.4.

- (20) a. second green ball (Matthei, 1982: 301)
b. big black balls (Hollebrandse & Roeper, 2014: 197)
- (21) a. tea pourer maker (Hollebrandse & Roeper, 2014: 179)
b. college student committee
- (22) a. Elmo’s sister’s ball (Pérez-Leroux et al., 2012: 301)
b. Taro-no otoosan-no kuruma (Japanese) (Terunuma et al., 2017: 627)
Taro-Gen father-Gen car
‘Taro’s father’s car’
- (23) a. tsukue-no osara-no ringo (Japanese) (Terunuma et al., 2017: 627)
table-Loc plate-Loc apple
‘an apple on the plate on the table’
b. zhuozi-shang de diezi-shang de pingguo (Mandarin Chinese)
table-Loc plate-Loc apple
‘an apple on the plate on the table’
- (24) the baby with the woman with the flowers (Pérez-Leroux et al., 2012: 302)
- (25) (Py=aida un=ati) daunaiur tyka-pa-uraz zyn kaiwada-pa-uraz kuwam
2SG=show 1SG=to guy see-PROG-REL girl wear-PROG-REL hat
‘(Show me) the guy that is seeing the girl that is wearing a hat’
(SG: singular; PROG: progressive tense; REL: a relational marker.) (Wapichana)
(Amaral & Leandro, 2018: 239)

Before the reanalysis, the definition of the 2-level tail-recursion sequence is stated as follows. The recursive rule “ $\alpha \rightarrow \langle \alpha B \rangle$ ” is potentially left-branching (i.e., left-expanding), for α in the brackets is on the left of B. The recursive rule “ $\beta \rightarrow \langle \beta A \rangle$ ” is also potentially left-branching. In contrast, recursive rules like “ $\alpha \rightarrow \langle B \alpha \rangle$ ” and “ $\beta \rightarrow \langle A \beta \rangle$ ” are right-branching. Left- or right-branching rules are called one-direction (left or right) tail-recursion rules. Besides, Rules like “ $\alpha \rightarrow \langle \alpha \beta \rangle$ ” and “ $\beta \rightarrow \langle \beta \alpha \rangle$ ” can be combined into the rule in (26), which further combines with (27) to produce the recursive rule in (28). Rules like (28) are (potentially) two-direction (left and

right) tail-recursion rules (cf. Rohrmeier et al., 2014). In sum, tail-recursion rules are a subtype of recursive rules that have the potentiality to generate new recursive sequences that recursively expand to the tail side(s) of the previous ones. When a recursive rule has two potential expanding directions, it is called a 2-direction recursive rule or an amphibious recursive rule. The two directions could be left and right, as in (28). They could also be left and inside, or right and inside, which will be exemplified in Section 2.4.

(26) $\alpha \rightarrow \langle \alpha \langle \beta \alpha \rangle \rangle$

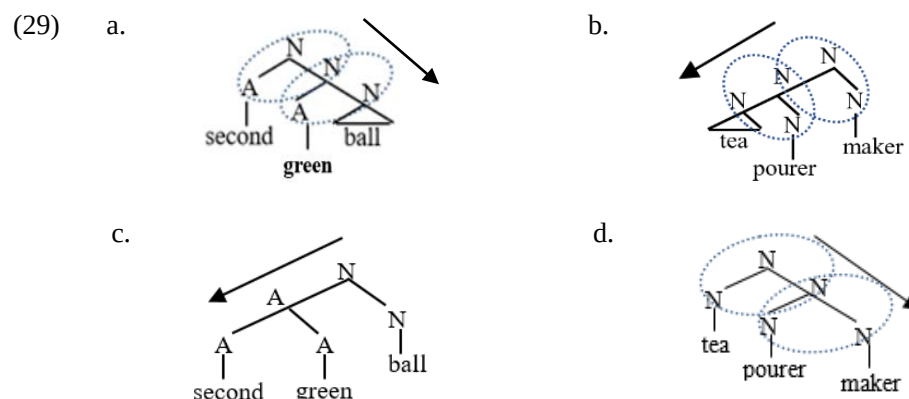
(27) $\beta \rightarrow M$

(28) $\alpha \rightarrow \langle \alpha \langle M \alpha \rangle \rangle$

To note, Rohrmeier et al. (2014: 73) also discuss the term “tail-recursion” and it seems that, in their opinion, “tail-recursion” is just coordination since they only list two expressions *The green, red, yellow, purple ... balls are in the box* and *This sentence continues and continues and continues* as the illustrations of “tail-recursion” in natural languages. However, the current paper considers tail-recursion as hierarchical/embedding.

Now, let’s reanalyze (20-24) one by one.

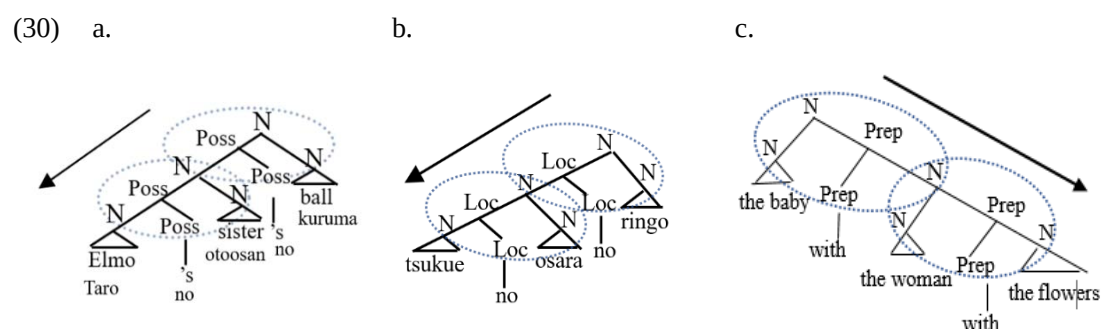
As shown in the tree diagram in (29a), the right-tail-recursion rule “ $N \rightarrow A N$ ” is used twice, resulting in “ $N \rightarrow A [A N]$ ”, with terminal words to generate 2-level tail-recursion sequences as in (20). As shown in (29b), the left-tail-recursion rule “ $N \rightarrow N N$ ” is used twice by way of “ $N \rightarrow [N N] N$ ” (not “ $N \rightarrow N [N N]$ ”), with terminals to generate 2-level tail-recursion sequences as in (21). The expressions in (20) and (21), matching (29a) and (29b), respectively, are different in expanding directions. Thus they are of different models of 2-level tail-recursion sequences. (Circles represent recursive rules in question. Arrows show expanding directions, which are in the top-down pattern.)



To note, apparently the expressions in both (20) and (21) have the head on the right with two modifiers. However, they are different in expanding directions and thus of different models. The reason is presented as followed. Presumably, we use a different tree diagram to make the spine in (29a) expand in the same direction as (29b), as shown in (29c). Then (29c) is not a 2-level recursive sequence since no rule is used twice in its derivation, and, moreover, (29c) is ungrammatical because there is no recursive rule like “ $A \rightarrow A A$ ” in English, and, also, *second green ball* cannot be interpreted as “second and green ball” as elaborated in Matthei (1982). In a similar line, presumably, we use a different tree model to make the spine in (29b) expand in the same direction as (29a), as shown in (29d). Then (29d) is different in meaning from (29b) since in (29d) the tail-recursion rule “ $N \rightarrow N N$ ” is used twice by way of “ $N \rightarrow N [N N]$ ” (not “ $N \rightarrow [N N] N$ ”) and, thus, the expression in (29d) means

“maker for tea”, different from “pourer for tea” in (29b).

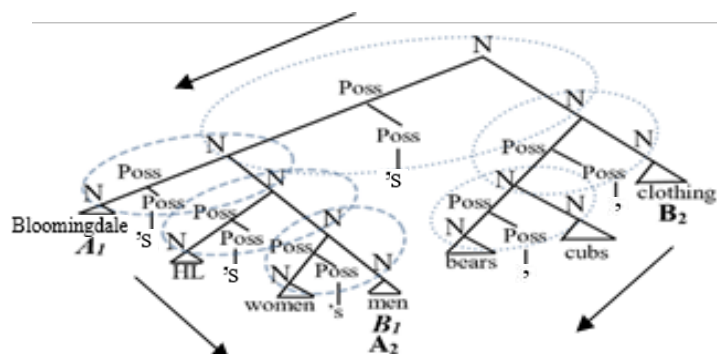
As shown in (30a), the (potentially) amphibious (left and right) tail-recursion rule “ $N \rightarrow N \text{ Poss } N$ ” (resulting from the combination of the two rules “ $N \rightarrow \text{Poss } N$ ” and “ $\text{Poss} \rightarrow N \text{ Poss}$ ”) is used twice by way of “ $N \rightarrow [N \text{ Poss } N] \text{ Poss } N$ ”, with terminal words to generate 2-level tail-recursion sequences like the ones in (22a–b). Similarly, as shown in (30b), the amphibious tail-recursion rule “ $N \rightarrow N \text{ Loc } N$ ” (resulting from the combination of “ $N \rightarrow \text{Loc } N$ ” and “ $\text{Loc} \rightarrow N \text{ Loc}$ ”) is used twice by way of “ $N \rightarrow [N \text{ Loc } N] \text{ Loc } N$ ”, with terminals to generate 2-level tail-recursion sequences like the ones in (23a–b). As shown in (30c), the amphibious tail-recursion rule “ $N \rightarrow N \text{ Prep } N$ ” (resulting from the combination of “ $N \rightarrow N \text{ Prep}$ ” and “ $\text{Prep} \rightarrow \text{Prep } N$ ”) is used twice by way of “ $N \rightarrow N \text{ Prep } [N \text{ Prep } N]$ ”, with terminals to generate 2-level tail-recursion sequences like the one in (24). The sequences in (22–24) (matching (30a–c), respectively) are different in expanding directions. Thus they are of different models.



It is noteworthy that the syntax of *John's sister's house* is analyzed as “ $[NP_{(n-1)}[NP_{(n)}[NP_{(n+1)}]]]$ ” in Martins & Fitch (2014: 22). The current paper reanalyzes it as “ $[[[NP_{(n-1)}]NP_{(n)}]NP_{(n+1)}]$ ”, in a line with the analyses of the expressions in (29). They are of different models, with different expanding directions, interfaced with different interpretations, with the former less acceptable.

It is also noteworthy that, amphibious tail-recursion rules like “ $N \rightarrow N \text{ Poss } N$ ” can be used repeatedly by way of “ $N \rightarrow [N \text{ Poss } N] \text{ Poss } [N \text{ Poss } N]$ ”, wherein some $N(s)$ can be further rewritten with “ $N \text{ Poss } N$ ”, with terminals to generate very complex tail-recursion sequences like the one in (31a/b). As shown in (31b), there are one long-distance dependency between A_1 and B_1 and one between A_2 and B_2 . Thus long-distance dependency exists in very complex tail-recursion sequences (cf. Munn, 1999; Li et al., 2019).

(31) a. Bloomingdale's [HL's women's men]'s [bears' cubs' clothing]
b.



2.4 To sub-classify some “category recursion” sequences like “recursive RCs” into 2-level tail-

nested-, and mixed-recursion sequences

The literature uses rather general terms like “Indirect Recursion”/“category recursion” to summarize the situations that some categories, say, RC, can be recursive. However, the following shows that “recursive RCs” are heterogeneous, which can be further sub-classified into different subtypes and various models.

2.4.1 The 2-level tail-recursion RC sequence

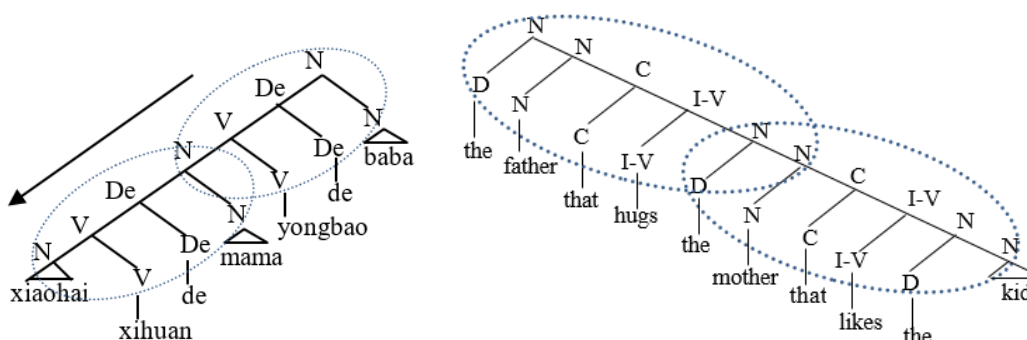
As shown in the tree diagram in (34a), the amphibious tail-recursion rule “ $N \rightarrow N V De N$ ” (resulting from the combination of “ $N \rightarrow De N$ ”, “ $De \rightarrow V De$ ”, and “ $V \rightarrow N V$ ”) specific to Mandarin Chinese can be used twice by way of “ $N \rightarrow [N V De N] V De N$ ”, with terminals to generate a Mandarin 2-level left-tail-recursion RC sequence like (32). However, as shown in (34b), there are 2-level right-tail-recursion RC sequences like (33) in English, resulting from the repeated use of the tail-recursion rule “ $N \rightarrow D N C I-V N$ ” by way of “ $N \rightarrow D N C I-V [D N C I-V N]$ ”, which is similar to the Wapichana RC sequence in (25) in Section 2.4.1, both right-tail-recursion.

(32) xiaohai xihuan De mama yongbao De baba
 kid like De mother hug De father
 ‘the father that the mother that the kid likes hugs’

(33) the father that hugs the mother that likes the kid

(34) a.

b.



The sequence in (32) is different from the one in (33) in that they are of different models with different expanding directions, as shown in (34a) and (34b). To note, (32)’s English translation *the father that the mother that the kid likes hugs* is a 2-level nested-recursion sequence (see Section 2.4.2), and (33)’s Chinese translation *yongbao xihuan haizi de mama de baba* is also a 2-level nested-recursion sequence (see Section 2.4.2).

2.4.2 The 2-level nested-recursion RC sequence

The left-tail-recursion rule “ $\alpha \rightarrow \langle \alpha B \rangle$ ” and the right-tail-recursion one “ $\alpha \rightarrow \langle A \alpha \rangle$ ” can be combined into a new, nested-recursion rule like (35), wherein α is nested in the middle of the other two terminal symbols A and B. Rules like the one in (35) can be repeatedly used to generate new sequences by way of recursively expanding to the “inside” of the previous ones. For example, the rule in (35) is used twice by way of (36) and then combines with recursion-terminating rules like “ $\alpha \rightarrow T$ ”, generating a 2-level nested-recursion sequence AATBB. The derivation of AATBB based on the rules in (35–36) hints a dependency shown with $A_1 A_2 T B_2 B_1$, wherein there are one long-distance dependency between A_1 and B_1 and one between A_2 and B_2 .

(35) $\alpha \rightarrow \langle A \alpha \rangle B$

(36) $\alpha \rightarrow \langle A \langle A \alpha \rangle B \rangle B$

As for natural language, the literature on this non-finite state model of nested-recursion sequences always involves RC, as illustrated in (37a–f) below. For example, Chomsky (1957: 22) argues that “it is clear that in English we can find a sequence $a + S1 + b$, where there is a dependency between a and b , and we can select as $S1$ another sequence containing $c + S2 + d$, where there is a dependency between c and d , then select as $S2$ another sequence of this form, etc.” and then presents the recursive rule in (37a) as an example of this non-finite state model of nested-recursion sequences. Chomsky & Miller (1963) further present the sequence in (37b) as an illustration of this model. To be more precise, (37c), part of (37b), is here reanalyzed as a 2-level nested-recursion RC sequence, which matches the tree diagram in (37e), wherein the (potentially) amphibious recursive rule “ $N \rightarrow N (C) N V$ ” (or “ $N \rightarrow N (C) N I-V$ ” to be more precise) is used twice by way of “ $N \rightarrow N (C) [N (C) N V] V$ ”, with terminal words to generate (37c), wherein *the rat* and *killed* are in long-distance dependency and so are *the cat* and *chase*. [To note, as mentioned in Section 2.3, a recursive rule is called an amphibious/2-direction one if it has two expanding directions. The two directions involved in the rule “ $N \rightarrow N (C) N V$ ” above is left and inside in that one of the N s is on the left tail side and the other is between the other symbols.] Also, as shown in the tree diagram in (37f), the amphibious recursion rule “ $N \rightarrow V N De N$ ” in Mandarin can be used twice by way of “ $N \rightarrow V [V N De N] De N$ ”, with terminal words to generate the Mandarin 2-level nested-recursion RC sequence in (37d), wherein *xihuan* ‘like’ and *mama* ‘mother’ are in long-distance dependency and so are *yongbao* ‘hug’ and *baba* ‘father’. (The two arrows in the opposite directions in (37e–f), among others, represent the (potential) “expanding to the inside”.)

(37) a. the man who said that S, is arriving today (Chomsky, 1957: 22)

b. The rat the cat the dog chased killed ate the malt. (Chomsky & Miller, 1963: 286)

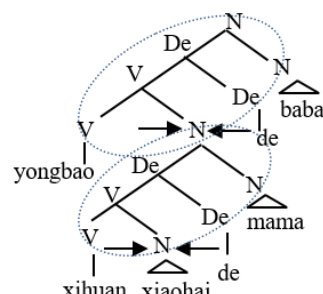
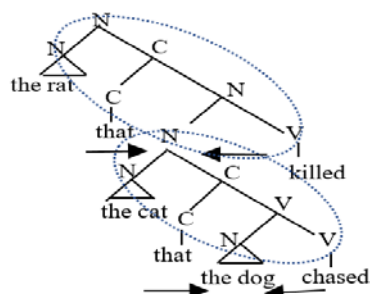
c. the rat (that) the cat (that) the dog chased killed

d. yongbao xihuan xiaohai De mama De baba

hug like kid De mother De father

‘the father that hugs the mother that likes the kid’

e. f.

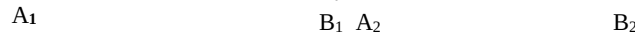


(37c)’s Chinese translation *gou zhui de mao shasi de laoshu* (similar to the sequence in (32) in Section 2.4.1) is a 2-level left-tail-recursion sequence, while (37d)’s English translation *the father that hugs the mother that likes the kid* (=33) is a 2-level right-tail-recursion sequence.

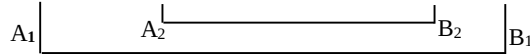
It is noteworthy that long-distance dependency exists in such complex 2-direction (left and right) tail-recursion sequences as mentioned in the analysis of the expression in (31a) in Section 2.4.1,

replicated as (38) below. It also exists in all nested-recursion sequences, e.g., the ones in (37), one of which is replicated as (39) below. Comparing (38) and (39), we can see that they are essentially different, for in the latter there is nestedness which the former lacks.

(38) a. Bloomingdale's [HL's women's men]'s [bears' cubs' clothing]



(39) the rat (that) [the cat (that) the dog chased] killed



2.4.3 The 2-level mixed-recursion RC sequences

A 2-level mixed-recursion sequence results from the combination of nested- and tail-recursion rules (thus called “mixed”), wherein there exists a recursive rule used twice (thus being 2-level). For example, the two tail-recursion rules “ $\alpha \rightarrow \langle \alpha B \rangle$ ” and “ $\alpha \rightarrow \langle A \alpha \rangle$ ” are combined into the nested-recursion rule “ $\alpha \rightarrow \langle \langle A \alpha \rangle B \rangle$ ”, which further combines with the rule “ $\alpha \rightarrow \langle A \alpha \rangle$ ” to generate a new, mixed-recursion rule, i.e., “ $\alpha \rightarrow \langle \langle A \langle A \alpha \rangle \rangle B \rangle$ ”, wherein there is a long-distance dependency hinted at with “ $\alpha \rightarrow \langle \langle A_1 \langle A \alpha \rangle \rangle B_1 \rangle$ ”. When this “ $\alpha \rightarrow \langle \langle A_1 \langle A \alpha \rangle \rangle B_1 \rangle$ ” combines with the recursion-terminating rule “ $\alpha \rightarrow T$ ”, a mixed-recursion sequence “ $A_1 A T B_1$ ” is generated. Along similar lines, more complex mixed-recursion sequences like $A_1 A A T B_1$, $A_1 A A A T B_1$ and $A_1 A_2 A A T B_2 B_1$ can be generated. For each of them, the rule “ $\alpha \rightarrow A \alpha$ ” is used at least twice. Mixed-recursion sequences in natural language are exemplified as follows, which are of various models.

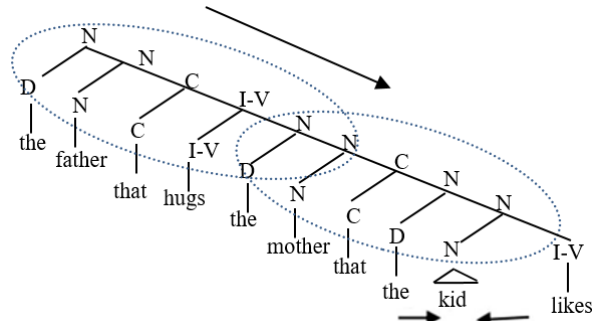
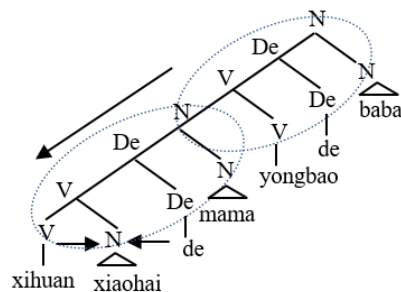
As shown in the tree diagram in (42a), in Mandarin, the amphibious (left and right) tail-recursion rule “ $N \rightarrow N \text{ V De } N$ ” is combined with the amphibious (right and inside) recursion rule “ $N \rightarrow V \text{ N De } N$ ” by way of “ $N \rightarrow [V \text{ N De } N] \text{ V De } N$ ”, with terminal words to generate a Mandarin 2-level mixed-recursion RC sequence in (40), wherein *xihuan* ‘like’ and *mama* ‘mother’ are in long-distance dependency. For it, there is one recursive rule, say, “ $N \rightarrow \text{De } N$ ” used twice. In a similar line, (41) is an English 2-level mixed-recursion RC sequence, matching the tree diagram in (42b).

(40) [xihuan xiaohai De mama] yongbao De baba
like kid De mother hug De father
'the father that the mother that likes the kid hugs'

(41) the father that hugs [the mother that the kid likes]

(42) a.

b.



As shown in (45a), the above two Mandarin rules “N→N V De N” and “N→V N De N” can also be combined by way of “N→V [N V De N] De N”, with terminal words to generate another 2-level mixed-recursion RC sequence in (43), wherein *yongbao* ‘hug’ and *baba* ‘father’ are in long-distance dependency. Similarly, as shown in (45b), (44) is an English 2-level mixed-recursion RC sequence,

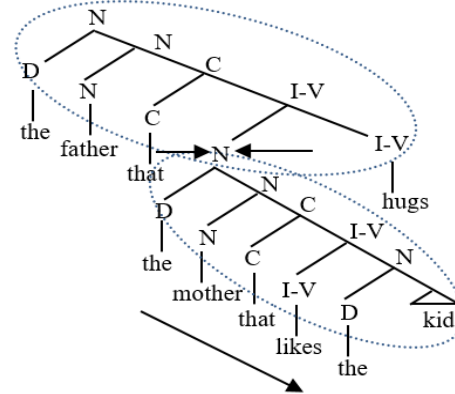
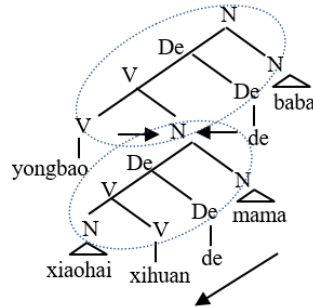
wherein *the father* and *hugs* are in long-distance dependency

(43) yongbao xiaohai xihuan De mama De baba
hug kid like De mother De father
'the father that hugs the mother that the kid likes'

(44) the father that [the mother that likes the kid] hugs

(45) a.

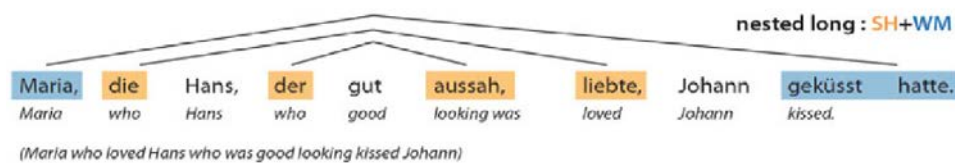
b.



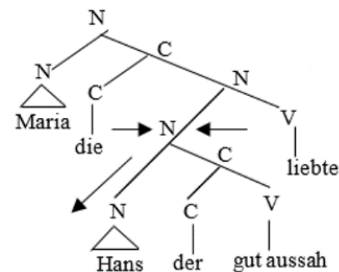
(40–45) are of different models of mixed-recursion RC sequences, with long-distance dependency, the distance in (43–45) longer than that in (40–42). In all of them there exists a recursive rule, e.g., “N→De N” in Mandarin or “N→N C” in English, used twice, and all result from the combination of nested and tail recursion. Thus all are 2-level mixed-recursion sequences. The expression in (40) and its English translation *the father that the mother that likes the kid hugs* (=44) are of different models of mixed recursion sequences; the same is true with the expression in (43) and its English translation *the father that hugs the mother that the kid likes* (=42).

To note, Friederici (2014) presents *Maria die Hans, der gut aussah liebte* in (46) as a German “nested” expression with two RCs (p.108). However, here in (47) is how it reanalyzed as a 2-level mixed-recursion RC sequence.

(46) Peter wusste, dass
(Peter knew, that)



(47)



2.5 The benefits of the proposed definition-classification system for recursion

The definition-classification system introduced in the current paper benefits theoretical and empirical work on recursion, as shown below.

Theoretically, it emphasizes the recursive nature of Merge, illustrates various kinds of non-finite

state models of recursive sequences within natural language, classifies them into tail, mixed and nested recursion sequences, and introduces the concept of the “levels” of recursive sequences. By doing this, it unifies some diverging definitions in the literature into a consistent definition-classification system and alleviate some confusion that exists in the field.

Empirically, the definition-classification system corroborates some experimental results in previous studies. For example, the distinction between long, 1-level recursive sequences like *Rose can eat a big apple* and short, 2-level recursive sequences like *(the) second green ball* collaborates the experimental results in Matthei (1982), which show that 5-year-olds cannot interpret short, 2-level recursive sequence like *(the) third red airplane* but can produce long, 1-level recursive sequences like *this is a tricky one, it's pose to be red, and you have to paint it*, as illustrated in (48).

(48) Child: E (5;7); Item 7, Experiment 1. (Matthei 1982: 317-318)

EM: Can you show me *the third red airplane*?

E: This is a tricky one. Let's see... It's yellow. [Pointing to the third airplane.] It's pose to be red. You have to pint it. Yeah. Paint it red.

Also, the sub-classification of the instances of “category recursion” into 2-level tail, mixed and nested recursion sequences corroborates the experimental results in the literature which show that children in general need 6-7 years to acquire 2-level tail recursion sequences, be they “recursive PPs, possessives, verbal nouns, complements or RCs” (Matthei, 1982; Pérez-Leroux et al., 2012; Sevcenco et al., 2015; Hollebrandse & Roeper, 2014; Yang et al., 2022) and also corroborates the experimental findings in Gibson (1998), Yang et al. (2022) and Friederici (2014), showing that “recursive RCs” are heterogeneous in that nested-, tail- and mixed-recursion RC sequences are acquired at different ages and involve different neurological activities in the brain.

3. Conclusion

Section 1 above introduces the problem that the current paper attempts to solve, namely, the lack of general consensus as to the definition of recursion. A number of linguistic phenomena have been described as recursive, including Merge (which is itself defined inconsistently), coordination and self-similar syntactic embedding (also called “Indirect Recursion” and “category recursion”). The article attempts to provide a formal definition-classification system for recursion in natural language, unifying some diverging definitions in the literature into a consistent system and alleviating some confusion that exists in the field. It is hopeful that, confronted with a given sequence, abiding by this formal definition-classification system, we can judge the level and subtype of the sequence and make comparisons between different sequences.

To note, the claim that recursion is the core of grammar in human natural language as mentioned in Section 1 does not necessarily mean that all subtypes and models of recursive sequences must be found in every language. The recursive sequences are, in principle, the explicit evidence of the recursive computation mechanism in human languages, and the absence of some recursive structures in some languages does not hurt the core status of recursion. There must be some level(s) and subtype(s) of recursive sequences in a language, be it 1-level, 2-level, etc. and be it tail-, mixed- or nested-recursion. It is quite probable that adults don't use 2-or-more-than-2-level nested-recursion sequences in daily life, as proven in Karlsson (2007), for reason of the “third factor” (which refers to “language-independent principles like structural architecture and computational efficiency”; see Chomsky, 2005: 8), even if such sequences are rather meaningful and grammatical.

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